

Semen quality of fertile US males in relation to their mothers' beef consumption during pregnancy.



BACKGROUND To look at possible long-term risks from anabolic steroids and other xenobiotics in beef, we examined men's semen quality in relation to their mother's self-reported beef consumption during pregnancy. **METHODS:** The study was carried out in five US cities between 1999 and 2005. We used regression analyses to examine semen parameters in 387 partners of pregnant women in relation to the amount of beef their mothers reported eating while pregnant. Mothers' beef consumption was also analysed in relation to the son's history of previous subfertility. **RESULTS** Sperm concentration was inversely related to mothers' beef meals per week ($P = 0.041$). In sons of "high beef consumers" (>7 beef meals/week), sperm concentration was 24.3% lower ($P = 0.014$) and the proportion of men with sperm concentration below $20 \times 10^6/\text{ml}$ was three times higher (17.7 versus 5.7%, $P = 0.002$) than in men whose mothers ate less beef. A history of previous subfertility was also more frequent among sons of "high beef consumers" ($P = 0.015$). Sperm concentration was not significantly related to mother's consumption of other meat or to the man's consumption of any meat. **CONCLUSIONS** These data suggest that maternal beef consumption, and possibly xenobiotics in beef, may alter a man's testicular development in utero and adversely affect his reproductive capacity.